

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

OPTICAL FIBER COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Due credit will be given to neatness.

1. Describe with the aid of ray diagram single mode step index fiber and multimode step index fiber. 10(CO2,3)

2.
 - (a) Explain the various types of dispersions occurring in optical fibers. Derive the expression for material dispersion parameter. 5(CO3)
 - (b) A graded index-fiber with a parabolic refractive index profile, supports a propagation of 742 guided modes, NA for air is 0.3, core radius is 70 μm . Determine :
 - (a) The wavelength of light propagating.
 - (b) Maximum diameter of fiber that gives single mode operation at the same mode of fiber. 5(CO2,3)

3.
 - (a) Explain the significance of 1.3 μm wavelength. Also, Explain Intramodal Dispersion. 5(CO4)
 - (b) A p-i-n photodiode has quantum efficiency of 0.5 at wavelength of 0.9 μm . Calculate :
 - (i) Received optical power if the mean photocurrent is 1 μA .
 - (ii) The corresponding number of received photons per second at this wavelength. 5(CO4)

4. (a) Define quantum Efficiency and Responsivity of a photo detectors. Derive the relation between them. 5(CO4)
- (b) The quantum efficiency of particular silicon RAPD is 80% for the detection of radiation at a wavelength of 0.9 micrometer. When the incident optical power is $0.5 \mu\text{W}$, the output current is $11 \mu\text{A}$. Determine the multiplication factor of the photodiode. 5(CO3,4)
5. (a) Describe with the neat diagram the back scatter method (OTDR) of attenuation measurement in optical fibre. 5(CO1,4)
- (b) Describe the cut-back method for measuring total transmission loss of fiber link. Explain why it is necessary to match the spot-size and numerical aperture of the incident beam and the fiber. 5(CO3)
6. (a) Explain the mechanism of WDM and Passive DWDM. 5(CO5)
- (b) Discuss Point to point link design and rise time budget. 5(CO5)

